FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
CONCEPT RECAPITULATION TEST – III
PAPER –2
TEST DATE: 26-04-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B

Sol. $\mu = iA$

$= i \cdot 12 \times \frac{a}{2} \times \left(\frac{2a}{\sqrt{3}}\right)$

$= \left(4\sqrt{3}a^2 i\right)$

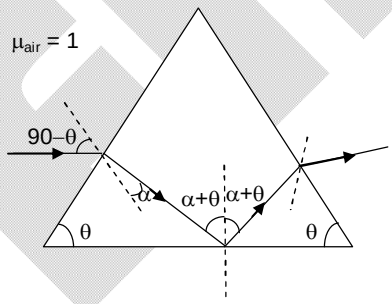
2. B

Sol. At base

$\alpha + \theta > \theta_C$

$\theta > \theta_C - \alpha$

At first refraction $\frac{\sin(90-\theta)}{\sin\alpha} = \frac{\mu g}{1}$



$\tan\theta > \frac{\mu_\omega - 1}{\sqrt{\mu_g^2 - \mu_\omega^2}}$

3. A

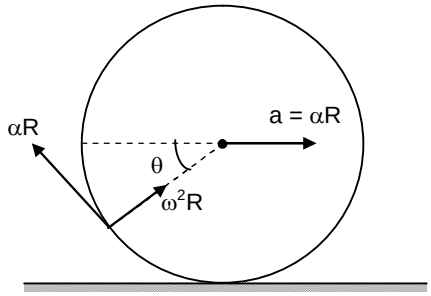
Sol. $v = v_0\left(1 - e^{-bt}\right)$

$P = F.\ v$

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4. B

Sol. $a_x = a + \omega^2 R\cos\theta - \alpha R\sin\theta$



$a_y = \omega^2 R\sin\theta + \alpha R\cos\theta$

5. BC

Sol. $r = \sqrt{(20-3t)^2 + (40-4t)^2}$

for r_{min}; $\frac{dr}{dt} = 0 \Rightarrow t = 8.8\,\text{sec}$

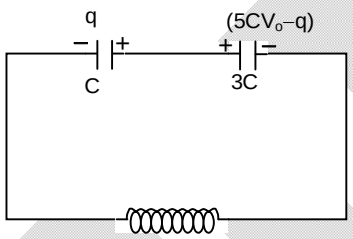
$r_{min} = 8\text{m}$

6. BC

Sol. Basic concept

7. ABCD

Sol. $-\frac{q}{c} - L\frac{di}{dt} + \frac{(5CV_0 - q)}{3C} = 0$



$\Rightarrow \quad q = \frac{5CV_0}{4} - \frac{9CV_0}{4}\cos\omega t \quad$ where $\omega = \frac{2}{\sqrt{3LC}}$

8. BC

Sol. $B2\pi r = \mu_0 \int \frac{J_0}{R^2} r(R-r)\cdot 2\pi r dr$

$B = \frac{J_0 \cdot r^2}{R^2}\left[\frac{R}{3} - \frac{r}{4}\right]$

9. BD

Sol. $E = -\frac{dV}{dx} = -6x^2$

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10. AC

Sol. $U = 2x^3 - 9x^2 + 12x$

$F = -6x + 18x - 12$

$F = 0 \Rightarrow x = 1,2$

Section – B

11. 0

Sol. $m_A v_0 = m_A v_1 + m_B v_2$

$\frac{1}{2}m_A v_0^2 = \frac{1}{2}m_A v_1^2 + \frac{1}{2}m_B v_2^2 + \frac{1}{2}kx^2$

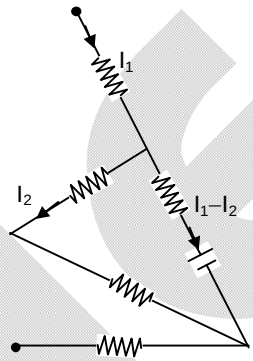
12. 0

Sol. $3I_1 - I_2 = 1$

$3I_2 - I_1 = 1$

$I_1 = 0.5, I_2 = 0.5$

$I_1 - I_2 = 0$



13. 4

Sol. $U = \frac{U_0}{2}\left[1 - \frac{x^2}{a^2}\right]^2$

$F = -\frac{dU}{dx} = \frac{2U_0 x}{a^2}\left(1 - \frac{x^2}{a^2}\right)$

Stable equation is at x = a

for x = a + y

$F = \frac{4U_0}{a^2}y\left(1 + \frac{y}{a}\right)$

14. 4

Sol. $\frac{d\ell}{\ell dT} = \alpha = \frac{T}{T_0}$

$\ln\left(\frac{\ell_2}{\ell_1}\right) = \frac{1}{T_0}\cdot\frac{T^2}{2}$

$T = \sqrt{T_0 \ln 4}$

15. 9

Sol. $\frac{Y\cdot\pi(2a)^2}{L}\times x = \frac{Y\cdot\pi(3a)^2}{L}\times(13-x)$

$4x = 9(13-x)$

$13x = 13\times 9$

$x = 9$

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16. 2

Sol. $I = I_0\left(1 - e^{-RT/L}\right)$

$$P_L = \frac{LdI}{dt} \cdot I = LI_0^2 \cdot \frac{R}{L} \cdot e^{-Rt/L} \cdot \left(1 - e^{-Rt/L}\right)$$

17. 4

Sol. $\frac{A_0}{\sqrt{2}} = A_0 \cdot e^{-10K}$

$A_1 = A_0 e^{-40K}$

18. 2

Sol. angular separation between two maximum $= \frac{\beta}{D} = \frac{\lambda}{d} = 10^{-3}$ rad

$\angle AOB = 0.12° = 2.08 \times 10^{-3}$ rad.

Total no. of maxima = 3

Hence minima here is 2.

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Chemistry

PART – II

Section – A

19. D
Sol. Theory based

20. D
Sol. $\Delta T = k_f \times \frac{\frac{1.45}{58}}{\frac{100}{1000}}$

$\Delta T = \frac{k_f}{4}$

$k_f = 4 \times 1.3 = 5.2$

$\Delta T = k_f \times m$

$2.6 = 5.2 \times \frac{\frac{7.81}{Mw}}{\frac{100}{1000}}$

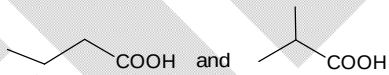
$Mw = 156.2\,g$

21. C
Sol. $0.04 \times \frac{200}{10} = 0.8\,mg$

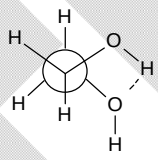
22. D
Sol.

I_3^-	Linear	sp^3d
SO_2	Bent	sp^2
O_3	Bent	sp^2
$\overset{\oplus}{N}O_2$	Linear	sp

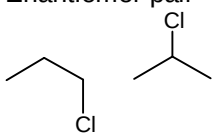
23. ABC
Sol.

(A) 

are chain isomers

(B) 
Gaurhe
(Ethyleneglycol) Intramolecular hydrogen bonding

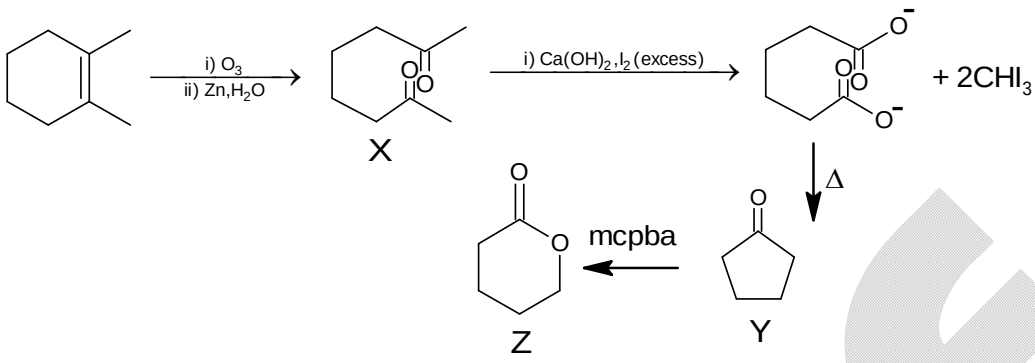
(C) Enantiomer pair

(D) 

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24. ACD

Sol.



25. D

Sol. $k = \text{time}^{-1}\ \text{mol}\,L^{-1}$ (zero order)

26. ABC

Sol. $10^{-13} = 0.001\times[OH]^2$

$[OH] = 10^{-5}$

$pOH = 5$

$$K_b = \frac{\left[\overset{+}{N}H_4\right]\left[\overset{\Theta}{O}H\right]}{[NH_4OH]}$$

$$2\times10^{-5}\ \frac{\left[\overset{+}{N}H_4\right]\left[10^{-5}\right]}{\left[10^{-2}\right]}$$

$$\left[\overset{+}{N}H_4\right] = 2\times10^{-2}M$$

$$= 0.02M$$

$$w_{NH_4Cl} = \frac{0.02}{2}\times53.5$$

$$= 0.535\ g$$

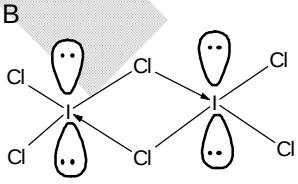
$$10^{-13} = \left[Mg^{2+}\right]\times\left[10^{-4}\right]^2$$

$$\left[Mg^{2+}\right] = 10^{-5}M$$

$$\frac{10^{-5}}{10^{-3}}\times100 = 1\%$$

27. B

Sol.



$Pb^{+4} + 2e \longrightarrow Pb^{+2}$

+2 oxidation state is more stable.

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28. BD

Sol. $n_{NaOH} = n_{H_2SO_4} \times \frac{1}{2}$

$n_{H_2SO_4} = 0.602$

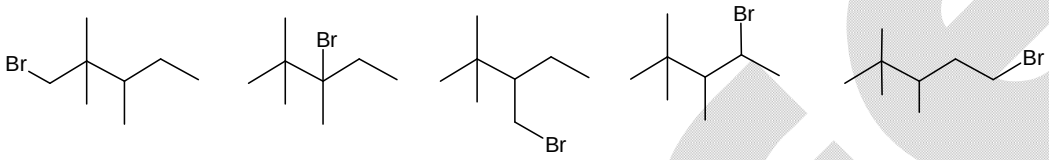
$w_{H_2SO_4} = 0.602 \times 98 = 59g(118\%)$

$w_{SO_3} = \frac{59-50}{18} \times 80 = 40\,g$

Section – B

29. 5

Sol.

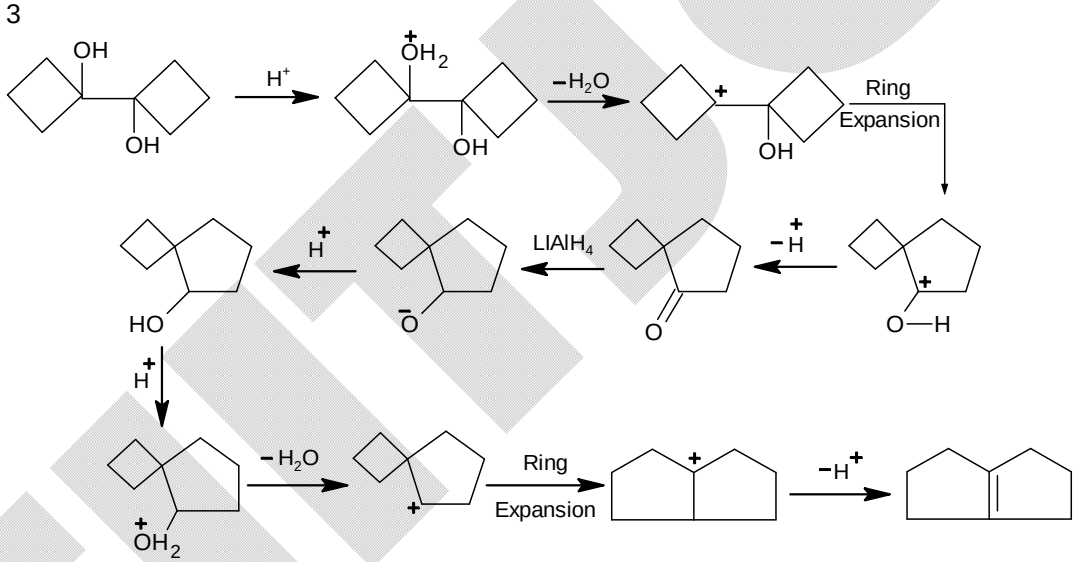


30. 3

Sol. $C_6H_{12}O_6 \xrightarrow{5HIO_4} 3HCOOH + CO_2 + 2HCHO$

31. 3

Sol.



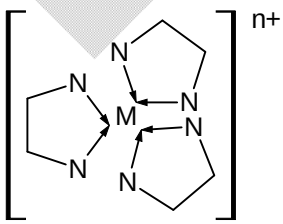
32. 6

Sol. $CH_3 - CHO \xrightarrow[ii)H_2O]{i)LiAlD_4} CH_3 - CHD(OH)$

Optically active

33. 3

Sol.



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34. 2

Sol. $r \propto \frac{1}{\sqrt{d}} \propto \frac{1}{\sqrt{Mw}}$

$\frac{r_{CH_4}}{r_{SO_2}} = \sqrt{\frac{64}{16}} = 2$

35. 5

Sol. $5 \rightarrow 4,\ 4 \rightarrow 2,\ 2 \rightarrow 1$

$5 \rightarrow 2$

$5 \rightarrow 1$

36. 7

Sol. $MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

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Mathematics

PART – III

Section – A

37. A

Sol. Let $I=\int_{-22}^{22}\frac{f(x)}{g(x)}dx$ ...(i)

$I=\int_{-22}^{22}\frac{f(-x)}{g(-x)}dx$ by using King's rule

$I=\int_{-22}^{22}\frac{f(-x)f(x)}{g(x)}dx$ ...(ii)

Adding (i) and (ii)

$2I=\int_{-22}^{22}\frac{f(x)(1+f(-x))}{g(x)}dx=\int_{-22}^{22}f(x)dx=2022$

38. C

Sol. Let $P(a\sec\theta,\ b\tan\theta)$ and $S(ae,0)$

Normal : $a\cos\theta x+b\cot\theta\, y=a^2+b^2$ $\quad G\left(\frac{a^2+b^2}{a\cos\theta},0\right)$

$SP=ex-a=ea\sec\theta-a$ and $SG=\frac{a^2+b^2}{a\cos\theta}-ae=e(ae\sec\theta-a)$

$\therefore\ \frac{SG}{SP}=e=4$

39. C

Sol. $A^2=A.A=ABA=AB=A$

and $B^2=BAB=BA=B$

$\therefore\ A=A^2=A^3=\cdots\ldots$ and $B=B^2=B^3=\cdots\ldots$

$\Rightarrow\left(A^{2022}+B^{2022}\right)^{2023}=(A+B)^{2023}$

$\because (A+B)^2=A^2+B^2+AB+BA=2(A+B)$

Similarly, $(A+B)^3=2^2(A+B)$ and so on

$\therefore\left(A^{2022}+B^{2022}\right)^{2023}=(A+B)^{2023}=2^{2022}(A+B)$

40. C

Sol. Let $\frac{m}{s}=\sin^2\theta$ and $\frac{n}{t}=\cos^2\theta$

$\therefore s+t=m\operatorname{cosec}^2+n\sec^2\theta$

$\because n\tan^2\theta+m\cot^2\theta\geq 2\sqrt{mn}$

$\therefore (s+t)_{min}=3+2\sqrt{mn}=3+2\sqrt{2}$

$\Rightarrow mn=2$ and $m=1, n=2$

Using $T=S_1\Rightarrow 2x+y=4$

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41. AC

Sol. Let $f(x) = ax^2 + bx + c$

Given, $-\frac{b}{2a} = 0 \Rightarrow b = 0 \Rightarrow f(x) = ax^2 + c$

$\Rightarrow f'(x) = 2ax \Rightarrow f''(x) = 2a$

Let g(x) has maximas at $x = \alpha \because g'(\alpha) = 0 \Rightarrow (f'(\alpha))^2 + f''(\alpha) = 0$ ...(i)

$\because g(\alpha) = |f'(\alpha)|e^{f(\alpha)} = 4\sqrt{e} \Rightarrow f'(\alpha) = 4$ and $f(\alpha) = \frac{1}{2}$

Now from $(f'(\alpha))^2 + f''(\alpha) = 0 \Rightarrow 16 + 2a = 0 \Rightarrow a = -8$

$f'(\alpha) = 4 \Rightarrow \alpha = -\frac{1}{4}$ and $f(\alpha) = a\alpha^2 + c = \frac{1}{2} \Rightarrow -\frac{8.1}{16} + c = \frac{1}{2} \Rightarrow c = 1$

$\therefore f(x) = -8x^2 + 1$ and $g(x) = |16x|e^{1-8x^2}$

42. ACD

Sol. Let $A(1,0,-1)$

The figure formed by the rod is cone.

Distance of A from plane (height of the cone): $h = \frac{1}{3}$

radius of base: $r = \sqrt{\ell^2 - 1} = \sqrt{3}$ $\therefore$ volume of cone $= \frac{1}{3}\pi r^2 h = \pi$

Centre of the circle: $\frac{x-1}{1} = \frac{y-0}{-2} = \frac{z+1}{2} = -\frac{1}{3}$ $\therefore$ centre $\left(\frac{2}{3}, \frac{2}{3}, -\frac{5}{3}\right)$

43. CD

Sol. (A) Let $f(x) = -(x-2)^{\frac{1}{3}}$ which is continuous decreasing function

$\Rightarrow \lim\limits_{h\to 0} \frac{f(2+h) - f(2)}{h}$ does not exist.

(B) Let $f(x) = e^{-x}$ which is continuous decreasing function

$\Rightarrow$ f(x) = 0 has no real solution

(C) Since f(x) is monotonically decreasing function and f(1 – x) is monotonically increasing function so y = f(x) and $y = f(1-x)$ meets at only point which is $\left(\frac{1}{2}, f\left(\frac{1}{2}\right)\right)$

$\therefore$ f(x) = f(1 – x) has only one solution $x = \frac{1}{2}$

44. ABD

Sol. Total number of matrices in $S = \frac{9!}{3!6!} = 84$

For number of non-singular matrices each row must have exactly one 1 and no two 1's must be present on the same column and this can be done in 6 ways. Hence, probability is $\frac{6}{84} = \frac{1}{14}$

Probability for identity matrix is $\frac{1}{84}$

For trace zero all three 1 must be placed at off diagonal places in ${}^6C_3$ ways. So required probability $= \frac{20}{84} = \frac{5}{21}$

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45. ABD

Sol. $f(x)=\frac{g^2(x)}{2}+3x^3-5$ given that $f'(x)>0$

$\therefore f'(x)>0 \Rightarrow g(x)g'(x)>-9x^2 \Rightarrow \int_0^1 g(x)d(g(x))>-9\int_0^1 x^2dx$

$\Rightarrow \frac{(g(1))^2-(g(0))^2}{2}>-3(1-0)\Rightarrow (g(1))^2>10$

Again, $\int_{-1}^0 g(x)d(g(x))>-9\int_{-1}^0 x^2dx \Rightarrow \frac{(g(0))^2-(g(-1))^2}{2}>-3(0-(-1))$

$\Rightarrow 16-(g(-1))^2>-6\Rightarrow (g(-1))^2<22$

$\because f(5)>f(0)\Rightarrow f(5)>3$

46. ACD

Sol. $3^x=4^{x^2}\Rightarrow x\log_2 3=x^2\log_2 4\Rightarrow x=0,\ x=\frac{1}{2}\log_2 3=\log_4 3$

Section – B

47. 5

Sol. $(\cot^{-1}x)(\tan^{-1}x)+\left(2-\frac{\pi}{2}\right)\cot^{-1}x-3\tan^{-1}x-3\left(2-\frac{\pi}{2}\right)>0$

$\Rightarrow(\cot^{-1}x)\left(\tan^{-1}x+2-\frac{\pi}{2}\right)-3\left(\tan^{-1}x+2-\frac{\pi}{2}\right)>0$

$\Rightarrow(\cot^{-1}x-3)(2-\cot^{-1}x)>0$

$\Rightarrow(\cot^{-1}x-3)(\cot^{-1}x-2)<0\Rightarrow 2<\cot^{-1}x<3$

$\Rightarrow x\in(\cot 3,\cot 2)$

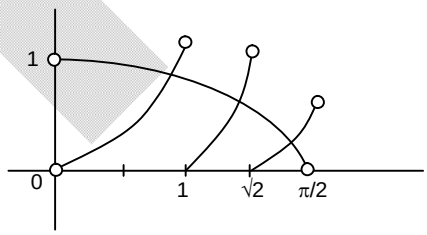
$\because \cot^{-1}x$ is decreasing function

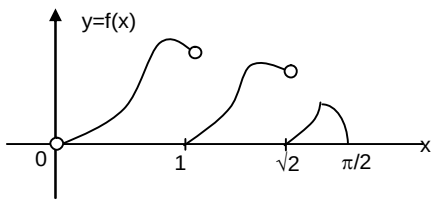
$\therefore \cot^{-1}a+\cot^{-1}b=\cot^{-1}\cot 3+\cot^{-1}\cot 2=5$

48. 7

Sol. $f(x)=\min\left(\{x^2\},\cos x\right)$ where $x\in\left(0,\frac{\pi}{2}\right)$

$$\{x^2\}=\begin{cases}x^2 & 0<x<1\\ x^2-1 & 1\le x<\sqrt{2}\\ x^2-2 & \sqrt{2}\le x<\frac{\pi}{2}\end{cases}$$





From graph discontinuous at 2 points and non-differentiable at 5 points.

$\therefore$ m + n = 7

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49. 2

Sol. $\int_0^x g'(x)\,dx = \int_0^x f(x)dx$

$\Rightarrow g(x) = \int_0^x f(x)dx$

$\Rightarrow \lim\limits_{x\to 0} xg\left(\frac{1}{x}\right)$ put $x = \frac{1}{t}$

$\Rightarrow \lim\limits_{t\to\infty} \frac{g(t)}{t} = \lim\limits_{t\to\infty} \frac{\int_0^t f(x)dx}{t}$

Let t = nT where T = 3 and $n \in I$

$\Rightarrow \lim\limits_{t\to\infty} \frac{\int_0^t f(x)dx}{t} = \lim\limits_{n\to\infty} \frac{\int_0^{3n} f(x)dx}{3n} = \lim\limits_{n\to\infty} \frac{n\int_0^3 f(x)dx}{3n} = \frac{6}{3} = 2$

50. 6

Sol. $a+b=(k-2)c$ and $ab=-(k-1)d$

$c+d=(k-2)a$ and $cd=-(k-1)b$

$\Rightarrow a((k-2)c-a)=-(k-1)((k-2)a-c)$

$\Rightarrow (k-2)ac-a^2=c(k-1)-(k-1)(k-2)a$ ...(i)

$\Rightarrow c((k-2)a-c)=-(k-1)((k-2)c-a)$

$\Rightarrow (k-2)ac-c^2=a(k-1)-(k-1)(k-2)c$ ...(ii)

Equation (i) – equation (ii)

$\Rightarrow c+a=(k-1)^2$

$\therefore a+b+c+d=100$

$\Rightarrow (k-2)(k-1)^2=100$

$\therefore k=6$

51. 3

Sol. Line $L: \frac{x-1}{1} = \frac{y-2}{2\sqrt{2}} = \frac{z}{0}$

$\because$ line L makes 45^0 with the given plane

$\therefore$ required distance PQ is perpendicular distance of point P from plane

$\Rightarrow PQ = \frac{\left|\sqrt{2}+7-\sqrt{2}-1\right|}{\sqrt{2+1+1}} = 3$

52. 4

Sol. $f(x)=x^3+ax^2+bx+c=0$ put $x=\frac{1}{t}$

We get, $ct^3+bt^2+at+1=0 \Rightarrow g(t)=0$

Hence, roots of f(x) and g(x) are reciprocal to each other.

$\therefore g\left(\frac{1}{3}\right)=0$ and $f\left(\frac{1}{4}\right)=0$

$\because g\left(\frac{1}{3}\right)=0$ and $\lim\limits_{x\to p}\frac{f(x)}{g(x)}$ exists

$\therefore f\left(\frac{1}{3}\right)=0$

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Hence roots of $x^3+ax^2+bx+c=0$ are $3,\frac{1}{3},\frac{1}{4}$

$\Rightarrow a=-\frac{43}{12};b=\frac{11}{6};c=-\frac{1}{4}$

$\Rightarrow \lim\limits_{x\to -1}\frac{f(x)+g(x)}{x+1}=3(c+1)-a-b=4$

53. 6

Sol. Normal at $P(x,y): Y-y=-\frac{1}{m}(X-m)\Rightarrow\left|\frac{x+my}{\sqrt{m^2+1}}\right|=y$

$\therefore \frac{dy}{dx}=\frac{y^2-x^2}{2xy}\Rightarrow (x-1)^2+y^2=1$

$\Rightarrow$ area bounded is π.

54. 1

Sol. Let A: event of drawing two same colour balls together

$P(A)=\frac{{}^3C_2+{}^nC_2}{{}^{n+3}C_2}=\frac{1}{2}\Rightarrow n^2-7n+6=0\Rightarrow n=1 \text{ or } 6$

Let B: event of drawing two same colour balls one by one

$P(B)=\frac{3}{n+3}.\frac{3}{n+3}+\frac{n}{n+3}.\frac{n}{n+3}=\frac{5}{8}$

$\Rightarrow n^2-10n+9=0$

$\Rightarrow n=1 \text{ or } 9$

$\therefore n=1$

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